## Purpose of policy

To provide a safe working environment for the patient, staff, and public and reduce waste disposal volumes and cost without compromising healthcare standards, and ensure compliance of all healthcare workers to the appropriate management of wastes, especially for those who are likely to be exposed to hazardous materials. As will provide standardized storage, handling, and proper disposal of waste.

## Definition

The waste disposal policy provides the method for handling, transporting, and disposing of medical waste to ensure cost reduction and safety for HCWs, sanitation workers, and the general public.

## Scope

This guideline is applicable to all healthcare workers, domestic, cleaning staff, and other workers including contracted service workers in healthcare facilities who handle or may come in contact with medical hazardous waste.

## Roles & Responsibilities

### A. All Patient Care Staff

- All staff should comply with this policy and report any hazards or incidents immediately to their supervisors and IP&C.
- Minimize the generation of waste
- Properly segregate and dispose of all materials
- Should be aware of the risks posed by handling

### B. Hospital management:

- Provide all required resources for the implementation of this policy

### C. Departmental Management

- enforce this policy.
- Education and training of staff on the safe disposal of waste

### D. Infection prevention and control

- Monitor compliance with policy during environmental audits.
- Participate in education and training regarding handling and disposal of waste.
- Provide feedback to the Health safety and environment committee

E. **Health, Safety, and Environment officer:**
   - Monitor Medical waste generation
   - Regular inspection to check proper segregation
   - Provide education and training regarding handling and disposal of waste.

F. **Housekeeping Staff:**
   - Responsible for the collection and transport of waste to the dumping area according to this policy.
   - Must be aware of the management of blood and body fluid spills.

## ❖ General concepts

- **Domestic waste (non-hazardous waste):**
  This includes items that would not release infectious materials and are not included in other hazardous categories e.g. paper, tissue paper, and Personal Protective Equipment (PPEs) not contaminated with blood or body fluids and do not come from isolation areas. As well, as soiled babies' napkins, incontinence pads, and sanitary pads/tampons. There is no evidence that hygienic waste poses a risk of infection to healthcare workers.

  **Recyclable waste:** Includes items for recycling such as cartoons, plastic cans**,** and glasses.

  **Food waste:** It includes waste foods during preparation and after each meal.

- **Medical (Hazardous) waste categories:**

  1- **Infectious waste:**

  - **Blood and blood products:** any liquid or semi-liquid blood or other potentially infectious materials including serum and plasma.
    - Sharps: include needles, scalpels, blades, broken glass, and pipettes.
    - **Pathological waste:** organs and body parts such as placentas and
    - amputated limbs
    - **Microbiological waste:** cultures and stocks of infectious agents from
    - laboratory work
    - **Contaminated items:** Items that would release blood or other potentially infectious materials in a liquid or semi-liquid state if compressed and are

capable of releasing these materials during handling.

- **Isolation rooms:** all waste generated in the isolation rooms are considered infectious waste including PPEs

**2- Non-infectious hazardous medical waste:**

- **Pharmaceutical:** Expired, unused, spilled, and contaminated pharmaceutical products, prescribed and proprietary drugs, vaccines, and sera that are no longer required.

- **Cytotoxic waste**: material that is, or maybe, is contaminated with a cytotoxic drug during the preparation, transport, or administration of chemotherapy.

- **Chemical waste** is generated by the use of chemicals in medical and laboratory procedures. Chemical wastes are any chemical that presents a physical or health hazard. For details refer to the chemical waste disposal procedure.

- **Radioactive:** Radioactive waste is materially contaminated with radio-isotopes, which arises from the medical or research use of radionuclides. It is produced, for example, during nuclear medicine imaging procedures.

1. **Generation of waste:**
   Minimization of waste product generation is a priority. It will save the hospital waste management and disposal costs.

2. **Segregation of waste:**
   Segregation of different types of waste is critical to the safe management of healthcare waste and helps control management costs refer to a color- the coding system in Appendix 1.

3. **Waste containers and color-coding**

**Color-coding:** The color-coded segregation system outlined in this section identifies and segregates waste on the basis of waste classification and suitability of treatment/disposal options in line with classifications mentioned on Appendex1.

**Waste containers:**

All designated infectious waste containers should have a biohazard label or be color-coded

**Container labeling:**

Each Container must have a label that identifies the type of waste and the name of the ward and the date of collection before disposal. This includes plastic bags and sharp boxes.

### 4. Disposal of liquid waste:

There are several ways to discard liquid infectious waste e.g. blood, body fluids, drain fluids, peritoneal dialysis fluids…. etc.

1- Disposal in the sewage system (sluice): Urine and body fluids and irrigation fluids during surgical procedures, peritoneal dialysis fluid

2- Suction canisters: The canisters are lined with plastic bags which should be sealed and discarded in the yellow bag

3- Blood units to be disposed of from the wards should be placed in a double yellow bag and transported to the blood bank for final disposal by incineration

4- For exchange blood transfusion in the pediatric ward, blood is collected in a 50 ml syringe and discarded in double yellow bags, and sealed well. The bag then should be sent to the blood bank for final disposal.

5- Highly contagious diseases (Category A) e.g. viral hemorrhagic fever (VHF):
Body fluids in the laboratory are chemically treated as per the VHF Handling guidelines HSM/OHSE/4. Body fluids generated in the wards can be discarded directly into the sewage system. If there is no toilet facility in the patient room, then the urine should be pretreated with 10,000ppm chlorine solution before disposing of in the sluice.

**Disposal of glass bottles:**

Must be segregated in separate dust bins or boxes labeled with (only glass bottles)

**Disposal of pharmaceutical waste:**

Pharmaceuticals will only be accepted by prior arrangement with be'ah after filling out the custom excels form.

**Disposal of Radioactive :**

**Protocol of solid radioactive waste disposal: (For more details please refer to Protocol of waste management in HOT lap and PET scan, Nuclear medicine ).**

Solid radioactive waste is segregated and disposed of into two waste forms.

1. Solid long half-life waste (> 90-day half-life) is disposed of appropriately and properly kept in their containers until returned to the supplier after working life.
2. Solid short half-life waste (< 90-day half-life) is disposed of into appropriate and properly labeled waste containers after 10 half-lives.
3. If long half-life radioisotopes cannot be separated from short half-life radioisotopes then the waste must always be disposed of into the long half-life waste container.

## 5. Collection of waste:

Housekeeping services should:

- Pick up waste at least once per day and as needed.
- Handle bags at the top so that the bags do not come in contact with the body.
- Do not use your hands to compress (squeeze) waste in containers/bags.
- Tie bags securely before placing them in a temporary holding area such as a dirty utility room.
- Do not store waste bags in hallways or corridors.
- Dispose of the sharp box when it is ¾th full.
- Fasten the cover of a full sharps container securely before removing.
- Decontaminate disposal bins/containers or frames when visibly soiled and cleaned them weekly with hospital-approved disinfectant.
- Decontaminate carts used daily to transport waste within the hospital using a hospital-approved disinfectant solution.
- Use leak-proof carts that are readily cleanable to transport infectious waste from the point of generation or storage to the point of disposal and treatment.
- Place yellow bags in a holding area for incineration.
- Pick up and discard broken glass using a mechanical device such as a forceps or a brush and dustpan.
- Broken glass should never be handled with gloved or non-gloved hands.
- Clean blood spills according to a written procedure (see “Blood Spills ” below).

## 6. Storage

- Storage facility at Royal hospital is outside the building, locked and a sign is posted at the entrance.
- There is a restriction on access to this area
- Wastes are stored in rigid wall containers yellow with biohazard signs for clinical waste and green for domestic waste.

## 7. Transport

- During the movement of wastes segregation must be maintained and the batch of waste managed according to the component with the highest level of risk.
- The staff should wear gloves during the transport of waste

### On-site transport

On-site transport of waste is usually from the initial storage point to an assembly storage or treatment area by means of trolleys, which have a solid base and bounding to contain spills. Trolleys used for the movement of clinical wastes are designed to prevent leakage, are easily cleaned, and minimize manual handling.
Waste collection rounds should be performed so as to minimize housekeeping hazards associated with the storage of waste at ‘user’ sites.

### Off-site transport

Relevant authorities must be contacted for special requirements for the transport of clinical wastes.

## 8. Final Disposal of waste:

<table>
<tr><th colspan="3">Final dispose of send to be'ah</th></tr>
<tr><th>Type of waste</th><th colspan="2">Disposal method</th></tr>
<tr><td>Domestic/ hygienic waste</td><td colspan="2">Landfill</td></tr>
<tr><td>Infectious</td><td>Autoclave</td><td rowspan="6">Shredding and exposing under 138C<br><br>3.8 bar for, 10Minutes</td></tr>
<tr><td>Microbiological cultures</td><td>Autoclaved inside a hospital and then incineration for the last disposal in be'ah</td></tr>
<tr><td>Sharps</td><td>Autoclave</td></tr>
<tr><td>Infectious waste contaminated with category A organisms such as viral hemorrhagic fever</td><td>Autoclave</td></tr>
<tr><td>Radioactive waste</td><td>After decay send for an autoclave</td></tr>
<tr><td>Cytotoxic</td><td>Incineration</td><td rowspan="2">Minimum 850C till becomes ash</td></tr>
<tr><td>Pathological waste</td><td>Small bats and tissue send for Incineration<br><br>Placenta, big tissue, and Organs send to a burial</td></tr>
<tr><td>Pharmaceutical waste:</td><td colspan="2">Will only be accepted by prior arrangement with be'ah after filling the custom excel form. And send for landfill or incineration depending to the categories of ems</td></tr>
</table>

## Management of spills:

Spill kits must be easily accessible at all work locations. For more details refer to biological spill management SOP.

**Occupational health and safety:**

The hospital must ensure that all waste is handled and disposed of safely. The waste management plan and procedures should be readily available to all workers involved.
All personnel who handle clinical and related waste must have access to information about:

- Occupational hazards of unprotected exposure;
- Policies and procedures for specific waste handling and prevention of injury and disease
- Personal protective equipment
- Immunization for Hepatitis B
- Access to medical care and counseling services in case of exposure to hazardous wastes
- Availability and access to first aid resources.

**Sharp/Needle sticks injury:**
Refer to the policy of management of needle stick injuries & incidents involving exposure to blood & body fluid POL/IPC/10/Vers.1.3

**Monitoring performance:**

**-** An audit should be performed semiannually to determine current performance in terms of safety, efficiency, environmental impact assessment, costs, and regulatory compliance
-All incidents and adverse events related to waste should be recorded. This is essential for identifying the factors that cause waste-handling injuries and determining the effectiveness of the waste management plan in reducing the incidence and severity of these injuries.

# Appendix 1 Color-coding

| Color | Description |
|---|---|
|  Domestic Hygienic waste (Black) | - Used to dispose of general hospital waste, paper, cardboard, cartoon, fabric<br>- Unbroken glass bottles<br>- Items that would not release (drip) blood or other potentially infectious materials in a liquid or semi-liquid state if squeezed.<br>- Hygienic waste including napkins and sanitary pads should be discarded in the black waste bag |
|  Infectious (Yellow) | - Used to dispose of infectious waste.<br>- Containers with blood/body fluids that cannot be emptied<br>- Items moderately or heavily soaked in blood or body fluids.<br>- Place infectious waste in the appropriate designated container, lined with yellow disposal bags.<br>- One garbage bin lined with a yellow disposal bag should be kept in the dirty utility room.<br>- Waste (specimens, cultures, and stocks of etiologic agents) generated from the microbiology laboratory should be put first in an autoclave bag and then after autoclaving be put in a yellow bag. |
|  Radioactive (Yellow) | - Radioactive wastes are discarded in the lead container designated for radioactive waste and discarded according to nuclear medicine guidelines. |
|  Cytotoxic (Purple) | Material that is, or maybe, be contaminated with a cytotoxic drug during the preparation, transport, or administration of chemotherapy.<br><br>- Chemotherapy should be disposed of in a purple bag assigned for cytotoxic waste |
|  Pathological (RED) | - Used for disposing of organs, body parts, and fetuses for burial and placenta.<br><br>The practice is to utilize the small yellow bag (size 42X60) for the body parts then be disposed of in the double red bag<br><br>For the placenta dispose of it in the double red bag |
|  Sharp box<br><br>For more information see | -<br>- Used to dispose of all needles, scalpels, pipettes, syringes, and broken glass items.<br>- Do not disassemble blades or needles from equipment.<br>- Discard sharps so that they do not protrude from the opening of the container. |

| | |
|---|---|
| (Guideline of management of sharp waste) | - Replace the sharp properly when the container is ¾full<br>- Containers should not be placed on floors, or on the lower shelves of trolleys kept where children might gain access.<br>- If carrying a sharp item, such as a syringe with needle, is unavoidable, then it must be carried in a container such as a kidney dish, so as to minimize the likelihood of a sharps injury.<br>- Sharps that are contaminated with cytotoxic or radioactive materials must be labeled and disposed of according to the procedures set out in the radiation protection and waste management policy |
| <br>Green transport container | - Large green containers transport domestic waste from the point of generation to the dumping area. These containers should be lockable. |
| <br>Yellow transport container | - Large yellow containers are used to transport clinical waste from point of generation to the dumping area. These containers should be lockable. |

## Appendix (2)
## Specifications of Trolleys and Bags

| Items | Specifications |
|---|---|
| **Sharps container** | 1. Yellow color<br>2. Puncture proof<br>3. Temporary lid closure while not in use<br>4. Permanent closure once locked.<br>5. Impermeable<br>6. Disposable- leak-proof<br>7. With a universal biohazard sign<br>8. Labeled SHARPS<br>9. With mark when in 3/4 ratio- preferable<br>10. Variable sizes ( e.g. 1,2, 5 , 7,10 L) that will fit trocar<br>**11. For radioactive:**<br>**12.** Special design container to fit the lead box. |
| **infectious waste bags** | 1. Yellow color bag – 150 microns<br>2. Leakproof<br>3. Chlorine free<br>4. With a universal biohazard symbol<br>5. *For highly infectious waste – that can withstand higher temperatures must be labeled<br>with the word “for autoclave “<br>6. *For other potentially infectious materials must be labeled with the word “infectious” |
| **Chemicals and Pharmaceutical waste box/containers** | 1. Brown color<br>2. For liquid waste<br>a. Metal container with cover<br>b. Plastic chemical-resistant containers with a cover for corrosive chemicals.<br>c. Re-using empty and cleaned original bottles or containers<br>3. Expired chemicals in original containers:<br>a. Brown box<br>4. Appropriate hazard symbol according to the hazard properties.<br>5. Label chemical bags and boxes with the word “chemical waste”<br>and symbol according to its hazardous properties using GHS<br>6. Pictograms.<br>7. For liquid waste- add more labels as stickers on its bottles with information<br>e.g. Chemical name and % composition.<br>8. *Special mercury containers with label word mercury |
| **Cytotoxic waste** | 1. Yellow Rigid puncture proof<br>2. Difficult to break open once locked<br>3. With **" cytotoxic symbol"** according to Globally Harmonized System<br>4. With opening on top to close and open |
| **Radioactive waste** | 1. Lead box with the radioactive symbol |
| **Pathological waste** | 1. Red color<br>2. Leak-proof bag<br>3. Thickness 150 microns above<br>4. With a " universal biohazard symbol"<br>5. Label with the word “ Deal with Islamic Fatwa” |

| | |
|---|---|
| **General waste container** | 150 microns |
| **Med Waste Bins** | 1. Color (according to the color of the bag e.g. yellow or brown)<br>2. Fully made of hard plastic or metal material<br>3. Chlorine free ( no phthalates / non-PVC) materials<br>4. Must be foot operated<br>5. Easy to clean and disinfect<br>6. No sharp edges<br>7. Easy to operate |
| **Transport Trolley** | **For infectious waste**<br>1. Yellow color<br>2. 2-wheeled trolley with cover<br>3. Rigid<br>4. Chlorine free ( no phthalates / non-PVC) materials<br>5. Made of hard plastic<br>6. With a "**<u>universal biohazard symbol"</u>**<br>7. Labeled with the word " healthcare waste or medical waste "<br>8. Easy to clean and disinfect<br>9. No sharp edges<br>10. Easy to maneuver<br><br>**For chemical waste**<br>1. Four-wheeled safety cart<br>2. Chemical resistance for corrosive chemicals<br>3. Prevents movement during transport<br>4. Easy to maneuver and stability<br>5. With brake system to prevent run away and cause the spill<br>6. No sharp edges<br>7. Easy to clean and decontaminate<br>A. Closed cart for flammable / toxic chemicals – meet OSHA and safety standard<br>B. Open cart to transport chemicals and pharmaceutical boxes |
| **PPE** | **Gloves**<br>1. Sharp resistant<br>2. Reusable<br>3. Has the ability to be cleaned and disinfected<br>**Mask**<br>1. Utility mask<br>**Apron**<br>1. Not easily torn<br>2. Waterproof<br>**Shoes**<br>1. Rubber boots<br>**Uniform**<br>1. Overall<br>2. Long sleeves |

| | |
|---|---|
| **Stickers /Label** | Information's<br>1. Name of Hospital<br>2. Department<br>3. Room number<br>4. Date |
| **Self-locking band** | Self-lock plastic band to seal the bags |
| **Spill kits**<br>**Biological**<br>**Corrosive chemicals**<br>**Formalin**<br>**Mercury**<br>**Cytotoxic** | Specific spill kits has different items but the list herein contains general specification:<br>1. Tools<br>2. PPE<br>3. Absorbent material<br>4. Neutralizer<br>5. Waste containers / bags |

**Note: Adopted from DEOH(Department of environment and Occupational Health department ,Ministry of Health**

## Appendix 3:
## Items which not allowed disposing in yellow bags

| List of Items | Remarks |
|---|---|
| 1. Gas cylinders, Aerosols or metallic sealed containers including inhalers and Unopened tinned goods. | 1, 2, 7 can come from anywhere |
| 2. Bottles of any type containing volatile or flammable liquids. | |
| 3.Body parts( to be dealt as per Islamic Jurisprudence) | 4, 5, 8, 9 & 10 are likely to come from pharmacies as expired drugs and Oncology center |
| 4. Flammable liquids. | |
| 5. Tinctures and compounds of iodine e.g. Povidone-iodine, Pyodine, Beta dine and Wok dine etc... must be identified and supplied separately. | 6 are likely to come from dental surgeries with attached laboratories |
| 6. Dental materials containing mercury or likely to contain low melting point metals. | 11 is only likely to come from nuclear medicine |
| 7. Batteries of any type. | |
| 8. Narcotic, psychotropic or regulated pharmaceuticals except with National Committee certificate and agreed process with MOH. | |
| 9. Chemical and pathology laboratory reagents. | |
| 10. Cytotoxic pharmaceuticals, sharps, giving sets, or empty containers. | |
| 11. These must be disposed of separately, notified, labeled, and color-coded as items for incineration only. | |
| 12. Radioactive materials unless decayed to 10x background. | |
| 13. Pharmaceuticals will only be accepted by prior arrangement with be'ah the following must also be segregated and separately notified as 'Implant for incineration only. | |
| 14. Artificial Devices & Prosthetics (The likeliest expected source of metal parts are when orthopedic implants or external fixings are removed in theatre) | |

## References

| Title of book/ journal/ articles/ Website | Author | Year of publication | Page |
|---|---|---|---|
| GCC Infection Prevention & Control Manual | GCC center for infection control | 2018 | 58-68 |
| Health Technical Memorandum 07-01 – Safe management of healthcare waste | Department of health UK | 2013 | |
| Guidelines for Environmental Infection Control in Health-Care Facilities | CDC | 2003 | |
| National guidelines for waste management in the health industry | National health and research medical council | 1999 | |

## Review / Revision History

| | Revision History | | |
|---|---|---|---|
| **Revision** | **Description of change** | **Author** | **Revision Date** |
| 1 | Initial Release | | 08/01/2026 |
| 2 | | | |
| 3 | | | |
| 4 | | | |
| 5 | | | |

| Written by | Checked by | Authorized by |
|---|---|---|
| Waria Alghafri<br>Amal AL Tai | Amal Al Tai | Director of infection Control and Occupational Health |